

EST100 - ENGINEERING MECHANICS

ANSWER KEY

PART A

1. State and prove Varignon's theorem of moments.

Varignon's theorem of moments states that the moment of a force about any axis is equal to the sum of moments of its components about that axis.

(OR) The algebraic sum of moments of a system of coplanar forces about a moment centre is equal to the moment of their resultant force about the same moment centre.

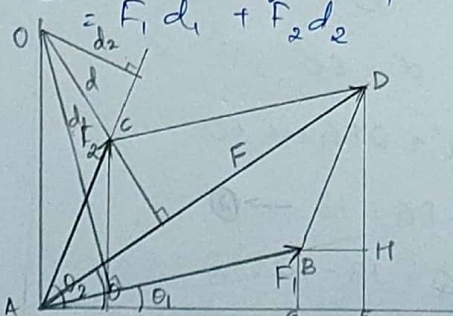
Proof:

Consider a force F acting at a point A . F_1 & F_2 are 2 components of F along any 2 directions.

Moment of F about an axis through an arbitrary point $O = F \times d$, where d is the arm of force F .

d_1 & d_2 are the arms of forces F_1 & F_2 respectively.

Sum of moments of components F_1 & F_2 about O



$$AE = AG + GE$$

$$= AG + BH$$

$$= AG + BD \cos \theta_2$$

$$= AG + AC \cos \theta_2$$

$$AD \cos \theta = AB \cos \theta_1 + AC \cos \theta_2$$

$$F \cos \theta = F_1 \cos \theta_1 + F_2 \cos \theta_2$$

or by OA,

$$F \times OA \cos \theta = F_1 \times OA \cos \theta_1 + F_2 \times OA \cos \theta_2$$

$$F \times d = F_1 \times d_1 + F_2 \times d_2$$

i.e., Moment of F about O = Moment of F_1 about O
+ Moment of F_2 about O .

i.e., Moment of a given force about any point
= Sum of moments of its components about that point

2. Find the magnitude of the two forces, such that if they act at right angles, their resultant is $\sqrt{10}$ N. But if they act at 60° , their resultant is $\sqrt{13}$ N.

$$\textcircled{1} R = \sqrt{10} \text{ N at } \alpha = 90^\circ$$

$$R^2 = P^2 + Q^2 + 2PQ \cos \alpha$$

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$$(\sqrt{10})^2 = P^2 + Q^2 + 2PQ \cos 90^\circ$$

$$P^2 + Q^2 = 10 \rightarrow \textcircled{1}$$

$$\textcircled{2} R = \sqrt{13} \text{ N at } \alpha = 60^\circ$$

$$(\sqrt{13})^2 = P^2 + Q^2 + 2PQ \cos 60^\circ$$

$$P^2 + Q^2 + PQ = 13$$

$$\text{From } \textcircled{1}, PQ = 13 - 10 = 3$$

$$(P+Q)^2 = P^2 + Q^2 + 2PQ$$

$$= 10 + 2 \times 3 = 16$$

$$P+Q = 4 \rightarrow \textcircled{a}$$

$$(P-Q)^2 = P^2 + Q^2 - 2PQ$$

$$= 10 - 2 \times 3 = 4$$

$$P-Q = 2 \rightarrow \textcircled{b}$$

Solving $\textcircled{a}$ & $\textcircled{b}$,

$$P+Q = 4$$

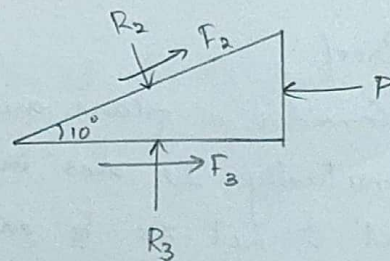
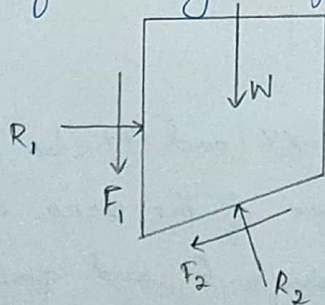
$$P-Q = 2$$

$$2P = 6$$

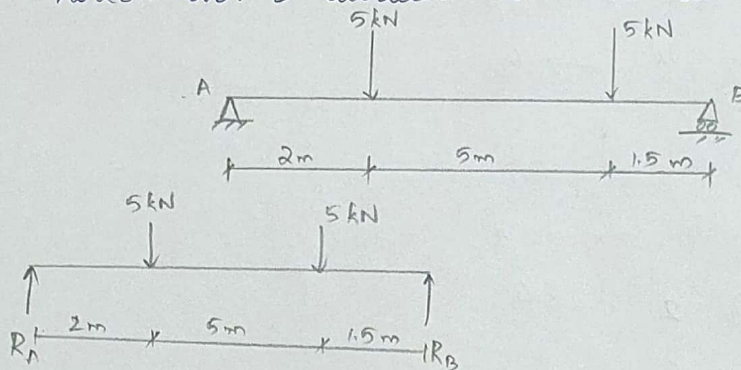
$$P = 3 \text{ N}$$

$$Q = 1 \text{ N}$$

3. A block weighing 1500 N, overlying a 10° wedge on a horizontal floor and leaning against a vertical wall, is to be raised by applying a horizontal force to the wedge. Assuming the coefficient of friction to the wedge. Draw the free body diagram of the wedge & the block.



4. A beam AB 8.5 m long is hinged at A and supported on rollers over B. Calculate reactions at A & B.



$$R_A + R_B = 10$$

$$\sum M_A = 0 \Rightarrow -R_B \times 8.5 + 5 \times 7 + 5 \times 2 = 0$$

$$R_B = 5.294 \text{ kN}$$

$$R_A = \underline{\underline{4.706 \text{ kN}}}$$

5. State & prove perpendicular axis theorem.

If I_{xx} and I_{yy} are the moment of inertia of an area A about mutually perpendicular axis xx and yy, in the plane of area, then the moment of inertia of the area about the ZZ axis which is perpendicular to xx and yy axis and passing through the point of intersection of xx and yy axis is given by

$$I_{zz} = I_{xx} + I_{yy}$$

Proof:

Consider a plane area A. Let xx and yy be the two mutually I_x axes in the plane of the area, intersecting at O. Let zz be an axis through O and perpendicular to the plane of area A.

Consider an elemental area dA at a distance y from the xx axis, x from the yy axis and r from the

ZZ axis, then

$$\begin{aligned} I_{zz} &= \int r^2 dA = \int (x^2 + y^2) dA \\ &= \int x^2 dA + \int y^2 dA \\ &= I_{xx} + I_{yy} \end{aligned}$$

$$\underline{I_{zz} = I_{xx} + I_{yy}}$$

6. A force of magnitude 44 N acts through a point $A(4, -1, 7)$ in the direction of vector $9i + 6j - 2k$. Find the moment of the force about the point $B(1, -3, 2)$.

Unit vector in the direction of vector $9i + 6j - 2k$ is

$$= \frac{9i + 6j - 2k}{\sqrt{9^2 + 6^2 + (-2)^2}} = \frac{9i + 6j - 2k}{11}$$

$$\text{Force } F = 44 \times \frac{9i + 6j - 2k}{11} = 36i + 24j - 8k$$

Position vector r of the point A w.r.t. point B,

$$\begin{aligned} r &= (4-1)i + (-1-(-3))j + (7-2)k \\ &= 3i + 2j + 5k \end{aligned}$$

$$M = r \times F = \begin{vmatrix} i & j & k \\ 3 & 2 & 5 \\ 36 & 24 & -8 \end{vmatrix}$$

$$= [(2 \times -8) - (5 \times 24)]i + [(5 \times 36) - (3 \times -8)]j + [(3 \times 24) - (2 \times 36)]k$$

$$= -136i + 204j + 0k$$

$$\underline{M = -136i + 204j}$$

7. The equations of motion of a particle moving in a straight line is given by $S = 20t + 5t^2 - 3t^3$ where S is in metres & t in seconds. Find the velocity & acceleration at the start.

$$S = 20t + 5t^2 - 3t^3$$

$$v = \frac{dS}{dt} = 20 + 10t - 9t^2$$

$$a = \frac{dv}{dt} = 10 - 18t$$

$$\text{At } t=0, v = 20 \text{ m/s}$$

$$a = \underline{\underline{10 \text{ m/s}^2}}$$

8. State and explain D'Alembert's principle.

D'Alembert's principle is an application of Newton's second law to a moving body.

Newton's law of motion $F = ma$ can be written as

$$F - ma = 0$$

The term $-ma$ is called inertia force and denoted by F_i . Inertia force is the resistance offered by a body to the change in its rest or uniform motion.

The magnitude of inertia force is equal to the product of mass & acceleration and it acts in a direction opposite to the direction of acceleration.

$$F - ma = 0$$

$$F + (-ma) = 0$$

$$\underline{F + F_i = 0}$$

The statement of the above equ. is known as D'Alembert's principle.

It states that the resultant of a system of force acting on a body in motion is in dynamic equilibrium with the inertia force.

9. A flywheel of mass 5 tonne starts from rest and gets up a speed of 150 rpm in 3 minutes. Find the average torque exerted on it if the radius of gyration of wheel is 50 cm.

$$m = 5 \text{ tonne} = 5 \times 10^3 \text{ kg}$$

$$\omega_0 = 0$$

$$N = 150 \text{ rpm}$$

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 150}{60} = 15.707 \text{ rad/s}$$

$$k = 50 \text{ cm}$$

$$t = 3 \text{ mins} = 180 \text{ s}$$

$$T = I \alpha$$

$$I = mk^2 = 5 \times 10^3 \times (50 \times 10^{-2})^2$$
$$= 1250 \text{ kgm}^2$$

$$\omega = \omega_0 + \alpha t$$

$$15.707 = 0 + \alpha \times 180$$

$$\alpha = 0.087 \text{ rad/s}^2$$

$$T = I \alpha = 1250 \times 0.087$$
$$= \underline{\underline{109.07 \text{ Nm}}}$$

10. Distinguish between simple harmonic motion and periodic motion.

Any motion which repeats after equal intervals of time is called a periodic motion.

Simple harmonic motion is a periodic motion. For a periodic motion to be simple harmonic, it should satisfy 2 conditions:

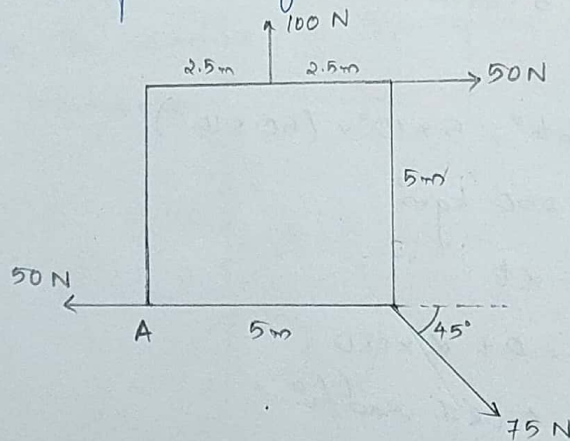
- i) Acceleration of the body performing periodic motion should be prop. to the distance of the body from a fixed point called centre of SHM or mean position of the body.
- ii) Acceleration of the body should always be directed towards the mean position.

PART B

MODULE 1

11.

- a) For the system of forces, determine the magnitude, direction & position of the resultant force about A.



$$\Sigma H = 50 + 75 \cos 45 - 50 = 53.033 \text{ N}$$

$$\Sigma V = 100 - 75 \sin 45 = 46.966 \text{ N}$$

Applying Varignon's Theorem of moments

$$R \cdot x = F_1 x_1 + F_2 x_2 + F_3 x_3 + F_4 x_4$$

$$= -100 \times 2.5 + 50 \times 5 + 75 \cos 45 \times 0 + 75 \sin 45 \times 5 - 50 \times 0 = 265.165 \text{ Nm}$$

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2} = \sqrt{53.033^2 + 46.966^2}$$

$$= 70.833 \text{ N}$$

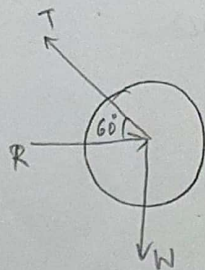
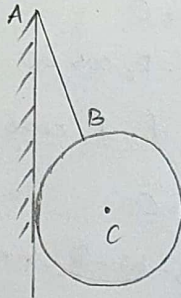
$$70.833 \times x = 265.165$$

$$x = \frac{265.165}{70.833} = \underline{\underline{3.743 \text{ m}}}$$

$$\tan \theta = \frac{\Sigma V}{\Sigma H} = \frac{46.966}{53.033} = 0.8856$$

$$\theta = \tan^{-1}(0.8856) = \underline{\underline{41.53^\circ}}$$

- b) A smooth sphere of weight W is supported by a string fastened to a point A on the smooth vertical wall, the other end is in contact with point B as shown in fig. If length of the string AC is equal to twice the radius of the sphere, find tension (T) in the string and reactions of the wall.



$$\Sigma H = 0$$

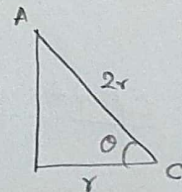
$$R - T \cos 60 = 0$$

$$R = T \cos 60 = \frac{T}{2}$$

$$\Sigma V = 0$$

$$T \sin 60 - W = 0$$

$$T \sin 60 = W$$



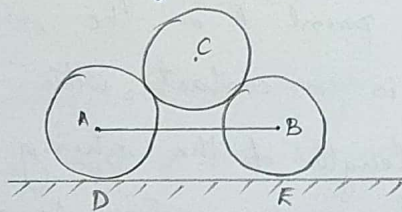
$$\cos \theta = \frac{r}{2r} = \frac{1}{2}$$

$$\theta = \underline{\underline{60^\circ}}$$

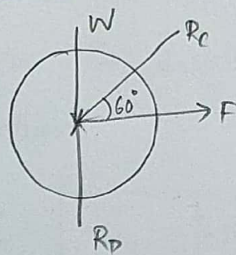
$$T = \frac{W}{\sin 60} = \underline{\underline{1.154 W}}$$

$$R = \frac{T}{2} = \frac{1.154 W}{2} = \underline{\underline{0.577 W}}$$

12. Two smooth ~~right~~ circular cylinders each of weight 50 N and radius 20 cm are connected at their centres by a string AB of length 40 cm and rest upon a horizontal plane as shown in fig. The cylinder above them has a weight of 100 N and radius 20 cm. Find the force in the string AB and pressure produced in the floor at the points of contact D and E.



Cylinder A



$$\Sigma H = 0$$

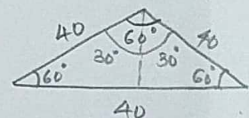
$$F - R_C \cos 60 = 0$$

$$F = R_C \cos 60$$

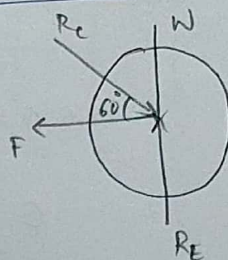
$$\Sigma V = 0$$

$$R_D - W - R_C \sin 60 = 0$$

$$R_D - R_C \sin 60 = 50$$



Cylinder B



$$\Sigma H = 0$$

$$-F + R_C \cos 60 = 0$$

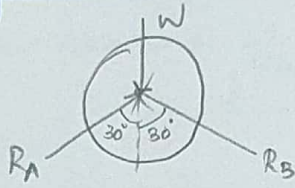
$$F = R_C \cos 60$$

$$\Sigma V = 0$$

$$-W + R_E - R_C \sin 60 = 0$$

$$R_E - R_C \sin 60 = 50$$

Cylinder C



$$\Sigma H = 0$$

$$R_A \sin 30 - R_B \sin 30 = 0$$

$$R_A = R_B$$

$$\Sigma V = 0$$

$$R_A \cos 30 + R_B \cos 30 - W = 0$$

$$2 R_A \cos 30 = 100$$

$$R_A = R_B = R_C = \underline{\underline{57.735 \text{ N}}}$$

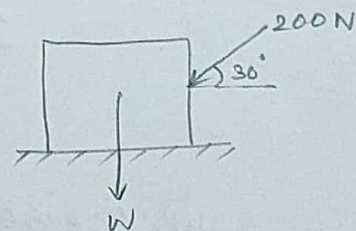
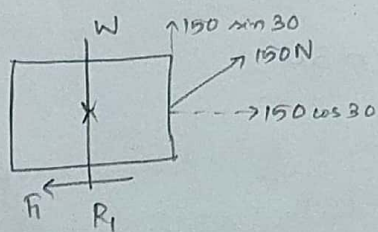
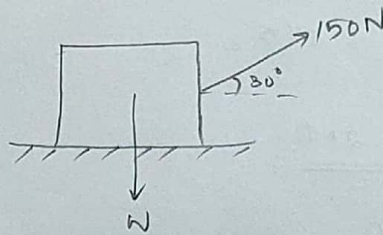
$$F = R_C \cos 60 = 28.86 \text{ N}$$

$$R_E = R_D = 50 + R_C \sin 60 = \underline{\underline{100 \text{ N}}}$$

MODULE 2

13.

- a) A body, resting on a rough horizontal plane, required a pull of 150 N inclined at 30° to the plane just to move it. It was found that a push of 200 N inclined at 30° to the plane just moved the body. Determine the weight of the body and the coefficient of friction.



$$\Sigma H = 0$$

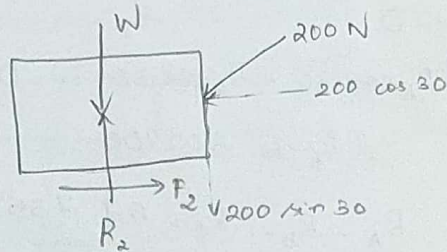
$$-F_1 + 150 \cos 30 = 0$$

$$F_1 = \underline{\underline{129.90 \text{ N}}}$$

$$\Sigma V = 0$$

$$R_1 - W + 150 \sin 30 = 0$$

$$W = R_1 + 150 \sin 30 \rightarrow (1)$$



$$\Sigma H = 0$$

$$F_2 - 200 \cos 30 = 0$$

$$F_2 = \underline{\underline{173.205 \text{ N}}}$$

$$\Sigma V = 0$$

$$R_2 - W - 200 \sin 30 = 0$$

$$(1) \Rightarrow R_2 - (R_1 + 150 \sin 30) - 200 \sin 30 = 0$$

$$R_2 - R_1 = 150 \sin 30 + 200 \sin 30 \rightarrow (2)$$

$$\mu = \frac{F}{R} \Rightarrow R_1 = \frac{F_1}{\mu}$$

$$\text{and } R_2 = \frac{F_2}{\mu}$$

$$(2) \Rightarrow \frac{173.205}{\mu} - \frac{129.904}{\mu} = 150 \sin 30 + 200 \sin 30$$

$$\frac{43.301}{\mu} = 175$$

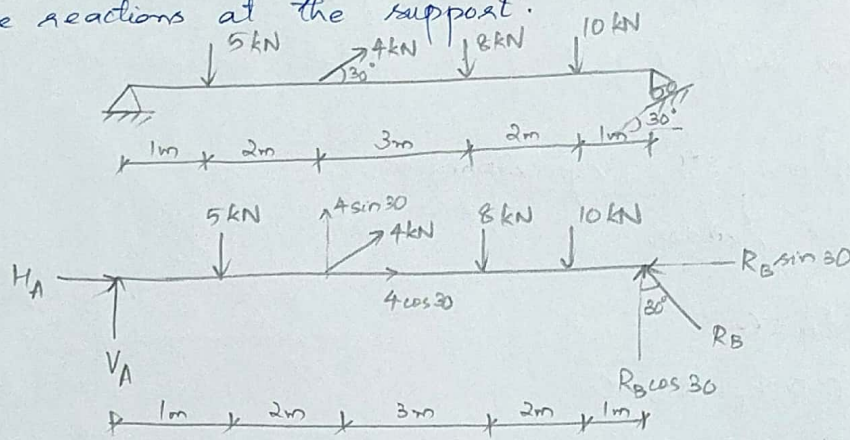
$$\mu = \underline{\underline{0.247}}$$

$$W = R_1 + 150 \sin 30$$

$$= 525.927 + 150 \sin 30$$

$$= \underline{\underline{600.92 \text{ N}}}$$

b) For the beam with loading shown in figure, determine the reactions at the support.



$$\sum V = 0$$

$$V_A - 5 + 4 \sin 30 - 8 - 10 + R_B \cos 30 = 0$$

$$V_A + R_B \cos 30 = 21 \rightarrow \textcircled{1}$$

$$\sum H = 0$$

$$H_A + 4 \cos 30 - R_B \sin 30 = 0$$

$$H_A - R_B \sin 30 = -3.464 \rightarrow \textcircled{2}$$

$$\sum M = 0$$

$$5 \times 1 - 4 \sin 30 \times 3 + 8 \times 6 + 10 \times 8 - R_B \cos 30 \times 9 = 0$$

$$R_B = 16.294 \text{ N}$$

$$\textcircled{2} \Rightarrow H_A = 4.683 \text{ N}$$

$$\textcircled{1} \Rightarrow V_A = 6.88 \text{ N}$$

$$R_A = \sqrt{H_A^2 + V_A^2} = \sqrt{4.683^2 + 6.88^2} \\ = 8.32 \text{ N}$$

14. A uniform ladder of 4 m length rests against a vertical wall with which it makes an angle of 45° . The coefficient of friction between the ladder and the wall is 0.4 and that between ladder and the floor is 0.5.

At what position along the ladder from the bottom end does the ladder slip, if a man whose weight is one-half of that of the ladder, ascends it.

Considering the equilibrium of the ladder

$$\sum F_H = 0$$

$$0.5 R_f - R_W = 0$$

$$R_f = 2 R_W$$

$$\sum F_V = 0$$

$$R_f - W - \frac{W}{2} + 0.4 R_W = 0$$

$$2 R_W - 1.5 W + 0.4 R_W = 0$$

$$2.4 R_W = 1.5 W$$

$$R_W = \frac{1.5 W}{2.4} = 0.625 W$$

$$\sum M = 0$$

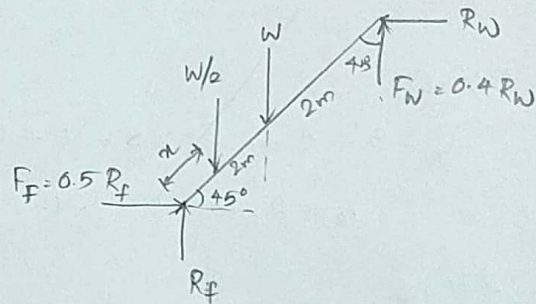
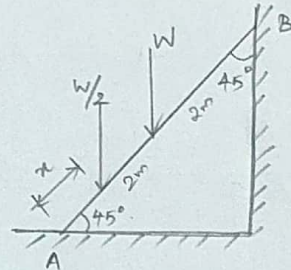
$$\frac{W}{2} \cdot x \cos 45 + W \times 2 \cos 45 - 0.4 R_W \times 4 \cos 45 - R_W \times 4 \sin 45 = 0$$

$$\frac{W}{2} \cdot x \cos 45 + W \times 2 \cos 45 - 0.4 \times 0.625 W \times 4 \cos 45 - 0.625 W \times 4 \sin 45 = 0$$

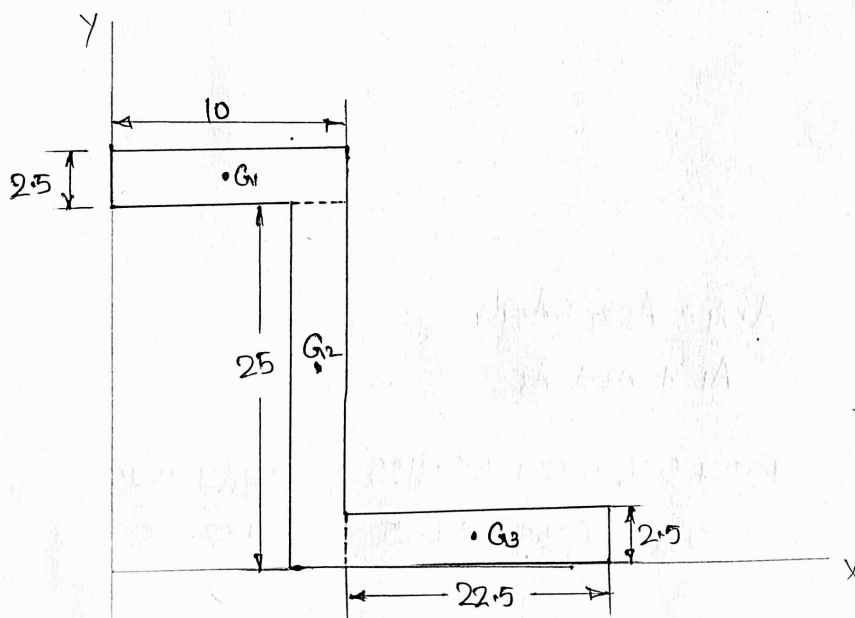
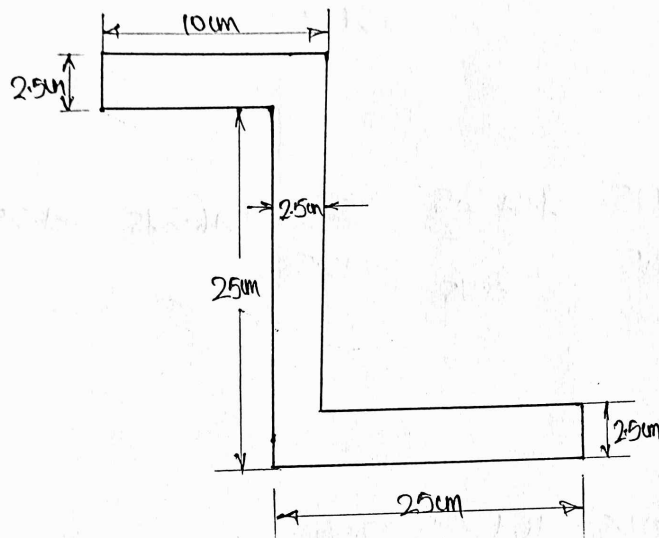
$$\frac{x}{2} + 2 - 0.4 \times 0.625 \times 4 - 0.625 \times 4 = 0$$

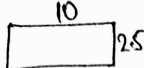
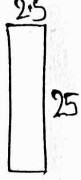
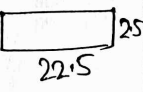
$$\frac{x}{2} = 1.5$$

$$x = 3m$$



15. Find the centroid of the cross-sectional area of a Z section shown in figure.



Sl No.	Fig	Area A_i	x_i	y_i	$A_i x_i$	$A_i y_i$
1.		2.5×10 $= 25$	$\frac{10}{2} = 5$	$25 + \frac{2.5}{2}$ $= 26.25$	125	656.25
2.		2.5×25 $= 62.5$	$7.5 + \frac{2.5}{2}$ $= 8.75$	$\frac{25}{2}$ $= 12.5$	546.875	781.25
3.		2.5×22.5 $= 56.25$	$10 + \frac{22.5}{2}$ $= 21.25$	$\frac{2.5}{2}$ $= 1.25$	1195.3125	70.3125

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3}$$

$$= \frac{125 + 546.875 + 1195.3125}{25 + 62.5 + 56.25} = \frac{1867.1875}{143.75} = \underline{\underline{12.989 \text{ cm}}}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3}$$

$$= \frac{656.25 + 781.25 + 70.3125}{25 + 62.5 + 56.25} = \frac{1507.8125}{143.75} = \underline{\underline{10.489 \text{ cm}}}$$

16. A force Q is directed from a point $A(4, 1, 4)$ meters towards a point $B(-3, 4, 1)$ meters. Determine the moment of force P about x and y axis, if it produces a moment of 1000 Nm about z axis.

Vector form of force $Q = |Q| \vec{\lambda}_{AB}$

$$\vec{Q} = |Q| \cdot \vec{\lambda}_{AB}$$

$$\vec{\lambda}_{AB} = \frac{(-3-4)\mathbf{i} + (4-1)\mathbf{j} + (1-4)\mathbf{k}}{\sqrt{(-7)^2 + 3^2 + (-3)^2}} = \frac{-7\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}}{\sqrt{67}}$$

$$\vec{Q} = |Q| \frac{-7\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}}{\sqrt{67}} = \frac{|Q|}{\sqrt{67}} (-7\mathbf{i} + 3\mathbf{j} - 3\mathbf{k})$$

Moment of force Q about the point origin,

$$\vec{M} = \vec{r} \times \vec{Q}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 1 & 4 \\ -7 & 3 & -3 \end{vmatrix} \frac{Q}{\sqrt{67}}$$

$$= \frac{Q}{\sqrt{67}} [(-3-12)\mathbf{i} - \mathbf{j}(-12-28) + \mathbf{k}(12-7)]$$

$$\vec{M} = \frac{|Q|}{\sqrt{67}} (-15\vec{i} - 16\vec{j} + 19\vec{k})$$

Moment about z axis is given as 1000 Nm,

$$\text{ie } \frac{|Q|}{\sqrt{67}} (19) = 1000$$

$$|Q| = \frac{1000 \times \sqrt{67}}{19} = 430.80 \text{ N}$$

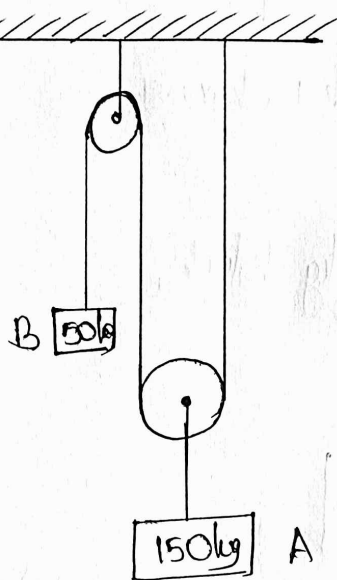
Moment about x axis

$$= \frac{430.80 \times -15}{\sqrt{67}} = \underline{\underline{-789.47 \text{ Nm}}}$$

Moment about y axis

$$= \frac{430.80 \times -16}{\sqrt{67}} = \underline{\underline{-842.10 \text{ Nm}}}$$

17. Determine the tension in the strings and accelerations of two blocks of mass 150kg and 50kg connected by a string and a frictionless and weightless pulley as shown in figure.



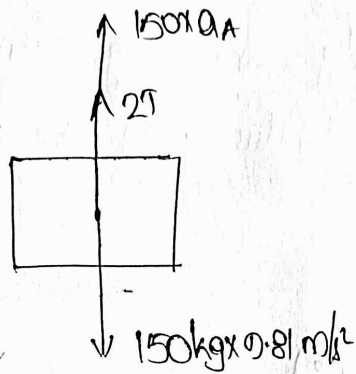
From the given figure, we can understand that 150kg block will move downwards and 50kg block will move upwards. The distance travelled by 150kg block will be half of distance travelled by 50kg block.

$$\therefore a_{50\text{kg}} = (a_{150\text{kg}}) \times 2$$

$$a_{150\text{kg}} = \frac{1}{2} a_{50\text{kg}}$$

$$a_A = \frac{1}{2} a_B$$

FBD of 150kg block,

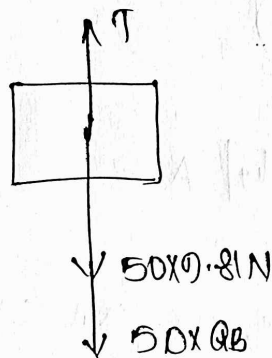


$$2T + 150QA = 150 \times 9.81$$

$$2T = 150 \times 9.81 - 150QA$$

$$T = 75 \times 9.81 - 75QA \quad \text{--- (1)}$$

FBD of 50kg block,



$$T = 50 \times 9.81 + 50QB \quad \text{--- (2)}$$

We know, $QA = \frac{1}{2}QB$

$$QB = 2QA$$

$$\therefore T = 50 \times 9.81 + 50 \times 2QA$$

$$T = 50 \times 9.81 + 100QA \quad \text{--- (3)}$$

Compare (1) and (3)

$$75 \times 9.81 - 75QA = 50 \times 9.81 + 100QA$$

$$25 \times 9.81 = 175QA$$

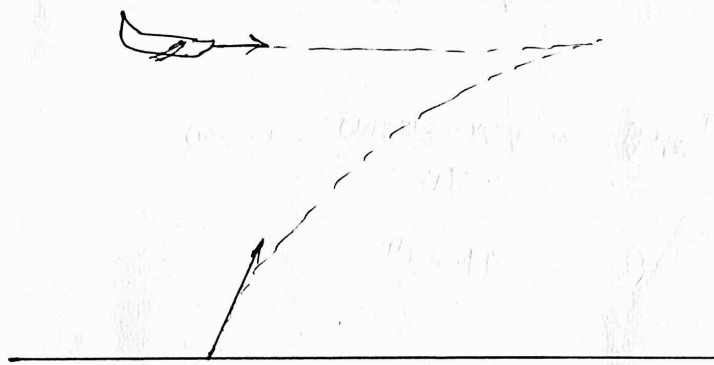
$$1.40 \text{ m/s}^2 = QA$$

$$QA = \underline{\underline{1.40 \text{ m/s}^2}}$$

$$QB = \underline{\underline{2.80 \text{ m/s}^2}}$$

$$T = 50 \times 9.81 + 100 \times 1.40 = \underline{\underline{630.5 \text{ N}}}$$

18. An aeroplane is flying at a height of 2000m with a horizontal velocity of 70 m/s. A shot is fired from a gun from the ground when aeroplane is exactly above the gun. What should be the minimum initial velocity of the shot and angle of elevation in order to hit the aeroplane.



The horizontal distance travelled by aeroplane and horizontal distance travelled by shot will be same.

$$\therefore v_x t = u \cos \theta \cdot t$$

$$70 \times t = u \cos \theta \cdot t$$

$$u \cos \theta = 70 \quad \text{--- (1)}$$

At the instant of hit the final velocity of shot become zero.

We know

$$v^2 = u^2 + 2as$$

Consider vertical component of velocity of shot,

$$0 = (u \sin \theta)^2 + 2gh \quad [\text{since } a = -g]$$

$$-u^2 \sin^2 \theta = -2gh$$

$$u \sin \theta = \sqrt{2gh} \quad \text{--- (2)}$$

$$\frac{(1)}{(2)} = \frac{u \cos \theta}{u \sin \theta} = \frac{70}{\sqrt{2 \times 9.81 \times 200}}$$

$$\tan \theta = \frac{\sqrt{2 \times 9.81 \times 200}}{70} = 0.89$$

$$\theta = \underline{\underline{41.82^\circ}}$$

From (1)

$$\Rightarrow u \cos \theta = 70$$

$$u = \frac{70}{\cos \theta} = \frac{70}{\cos 41.82}$$

$$u = \underline{\underline{93.95 \text{ m/s}}}$$

12. A flywheel rotates for 5s with a constant angular acceleration and describes during that time 100rad. Then it rotates with constant angular velocity and during the next 5s describes 80rad. Find the initial angular velocity and angular acceleration.

Case I (const. angular acceleration)

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

Case II (const. angular velocity)

$$\theta = \omega t$$

$$80 = \omega \times 5$$

$$\omega = \frac{80}{5} = \underline{\underline{16 \text{ rad/s}}}$$

Given:-

$$t = 5s$$

$$\theta = 80 \text{ rad}$$

$\omega = 16 \text{ rad/s}$ in case 2, is constant angular velocity.

The final angular velocity of case I will be the angular velocity of case 2.

So in case I,

$$\omega = 16 \text{ rad/s}$$

$$\theta = 100 \text{ rad}$$

$$t = 5s$$

By using the equations,

$$(1) \Rightarrow \omega = \omega_0 + \alpha t \Rightarrow 16 = \omega_0 + 5\alpha \quad \text{--- (1)}$$

$$(2) \Rightarrow \theta = \omega_0 t + \frac{1}{2} \alpha t^2 \Rightarrow 100 = \omega_0 \times 5 + \frac{1}{2} \times \alpha \times 25$$

$$\Rightarrow 20 = \omega_0 + 2.5\alpha \quad \text{--- (2)}$$

$$\Rightarrow 20 = (16 - 5\alpha) + 2.5\alpha \quad [\text{from eq. (1)}]$$

$$\Rightarrow 20 = 16 - 5\alpha + 2.5\alpha$$

$$4 = -2.5\alpha$$

$$\alpha = \frac{-4}{2.5} = -1.6 \text{ rad/s}^2$$

$$\alpha = \underline{\underline{-1.6 \text{ rad/s}^2}}$$

$$(1) \Rightarrow 16 = \omega_0 + 5\alpha$$

$$16 = \omega_0 + 5 \times -1.6$$

$$\omega_0 = 16 + 8$$

$$\omega_0 = \underline{\underline{24 \text{ rad/s}}}$$

20. A flywheel is made of steel ring, 30mm thick and 300mm wide plate, with mean diameters of 1.5m. If initially the flywheel is rotating at 250rpm, find the time taken by the wheel in coming to rest due to frictional couple of 150Nm. Take mass density of steel as 7900 kg/m^3 . Neglect the effect of spokes.

Given:-

Thickness, $h = 30 \text{ mm} = 30 \times 10^{-3} \text{ m}$

Width, $w = 300 \text{ mm} = 300 \times 10^{-3} \text{ m} = 0.3 \text{ m}$

$d_m = 1.5 \text{ m}$

$r_m = 0.75 \text{ m}$

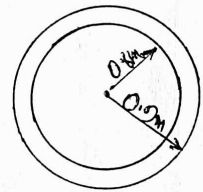
$N_0 = 250 \text{ rpm}$

$\omega_0 = \frac{2\pi \times 250}{60} = \frac{25\pi}{3} \text{ rad/s}$

$\omega = 0$

$T = 150 \text{ Nm}$

Density, $\rho = 7900 \text{ kg/m}^3$



Mass of flywheel = density $\times$ Volume

Volume of ring = $\pi R^2 h - \pi r^2 h$

$$\text{Volume, } V = \pi h [R^2 - r^2]$$

$$= \pi \times 30 \times 10^{-3} [0.7^2 - 0.6^2]$$

$$= 0.0424 \text{ m}^3$$

$$\text{Mass of flywheel} = \rho \times V$$

$$= 7700 \times 0.0424$$

$$= 335.05 \text{ kg}$$

Mass moment of inertia,

$$I = m r_m^2$$

$$= 335.05 \times 0.75^2$$

$$= 188.46 \text{ kgm}^2$$

We know,

$$\tau = I \alpha$$

$$150 = 188.46 \times \alpha$$

$$\alpha = \underline{\underline{0.796 \text{ rad/s}^2}}$$

$$\omega = \omega_0 + \alpha t$$

$$0 = \frac{25\pi}{3} + 0.796 \times t$$

$$t = \frac{-26.17}{-0.796} = \underline{\underline{32.88 \text{ s}}}$$